



**ROYAL SCHOOL OF APPLIED & PURE SCIENCES  
(RSAPS)**

**DEPARTMENT OF CHEMISTRY**

**COURSE STRUCTURE & SYLLABUS  
(BASED ON NATIONAL EDUCATION POLICY 2020)**

**FOR  
M.Sc. CHEMISTRY**

**AY 2026-27**

## STRUCTURE OF THE SYLLABUS FOR 2 YEAR PG PROGRAMME

SCHOOL NAME - ROYAL SCHOOL OF APPLIED AND PURE SCIENCES (RSAPS)

DEPARTMENT NAME - CHEMISTRY

PROGRAMME NAME - M.Sc. in CHEMISTRY

| 1 <sup>st</sup> SEMESTER                                                                                 |                                                              |       |            |       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------|------------|-------|
| COURSE CODE                                                                                              | COURSE TITLE                                                 | LEVEL | CREDIT     | L-T-P |
| CHY014C101                                                                                               | Physical Chemistry I                                         | 400   | 4          | 4-0-0 |
| CHY014C102                                                                                               | Organic Chemistry I                                          | 400   | 4          | 4-0-0 |
| CHY014C103                                                                                               | Inorganic Chemistry I                                        | 400   | 4          | 4-0-0 |
| CHY014C104                                                                                               | Quantum Chemistry                                            | 400   | 4          | 4-0-0 |
| CHY014C115                                                                                               | Inorganic Chemistry Lab                                      | 400   | 4          | 0-0-8 |
| SWAYAM CODE 1                                                                                            | Swayam 1                                                     | 400   | 3/4/5      |       |
| TOTAL CREDIT FOR 1 <sup>st</sup> SEMESTER                                                                |                                                              |       | 20 + 3/4/5 |       |
| 2 <sup>nd</sup> SEMESTER                                                                                 |                                                              |       |            |       |
| COURSE CODE                                                                                              | COURSE TITLE                                                 | LEVEL | CREDIT     | L-T-P |
| CHY014C201                                                                                               | Physical Chemistry II                                        | 400   | 4          | 4-0-0 |
| CHY014C202                                                                                               | Organic Chemistry II                                         | 400   | 4          | 4-0-0 |
| CHY014C203                                                                                               | Inorganic Chemistry II                                       | 400   | 4          | 4-0-0 |
| CHY014C204                                                                                               | Analytical Chemistry                                         | 400   | 4          | 4-0-0 |
| CHY014C215                                                                                               | Organic Chemistry Lab                                        | 400   | 4          | 0-0-8 |
| SWAYAM CODE 2                                                                                            | Swayam 2                                                     | 400   | 3/4/5      |       |
| TOTAL CREDIT FOR 2 <sup>nd</sup> SEMESTER                                                                |                                                              |       | 20 + 3/4/5 |       |
| TOTAL CREDIT FOR 1 <sup>st</sup> YEAR = 40 + 6/8/10                                                      |                                                              |       |            |       |
| 3 <sup>rd</sup> SEMESTER                                                                                 |                                                              |       |            |       |
| COURSE CODE                                                                                              | COURSE TITLE                                                 | LEVEL | CREDIT     | L-T-P |
| CHY014C301                                                                                               | Spectroscopy I                                               | 500   | 4          | 4-0-0 |
| CHY014C302                                                                                               | Biochemistry & Bioinorganic Chemistry                        | 500   | 4          | 4-0-0 |
| CHY014C303                                                                                               | Environmental & Green Chemistry                              | 500   | 4          | 4-0-0 |
| CHY014C314                                                                                               | Physical Chemistry Lab                                       | 500   | 4          | 0-0-8 |
| CHY014C325                                                                                               | Project -I                                                   | 500   | 8          | 0-0-0 |
| SWAYAM CODE 3                                                                                            | Swayam 3                                                     | 500   | 3/4/5      |       |
| TOTAL CREDIT FOR 3 <sup>rd</sup> SEMESTER                                                                |                                                              |       | 24 + 3/4/5 |       |
| OR 3 <sup>rd</sup> SEMESTER<br>(For students with 3 <sup>rd</sup> and 4 <sup>th</sup> Semester Research) |                                                              |       |            |       |
| CHY014C321                                                                                               | RESEARCH PROJECT – PHASE I                                   | 500   | 24 + 3/4/5 | 0-0-0 |
| 4 <sup>th</sup> SEMESTER                                                                                 |                                                              |       |            |       |
| COURSE CODE                                                                                              | COURSE TITLE                                                 | LEVEL | CREDIT     | L-T-P |
|                                                                                                          | Dissertation (students with research in 4 <sup>th</sup> Sem) |       |            |       |
| (for 'Coursework only' in lieu of Research)                                                              |                                                              |       |            |       |
| CHY014C401                                                                                               | Spectroscopy II                                              | 500   | 4          | 4-0-0 |
| CHY014C402                                                                                               | Chemical Kinetics & Catalysis                                | 500   | 4          | 4-0-0 |
| CHY014C403                                                                                               | Heterocyclic Compounds & Medicinal Chemistry                 | 500   | 4          | 4-0-0 |
| CHY014C404                                                                                               | Organometallic Chemistry & Catalysis                         | 500   | 4          | 4-0-0 |
| CHY014C426                                                                                               | Project -II                                                  | 500   | 12         | 0-0-0 |
| SWAYAM CODE 4                                                                                            | Swayam 4                                                     | 500   | 3/4/5      |       |
| OR 4 <sup>th</sup> SEMESTER<br>(For students with 3 <sup>rd</sup> and 4 <sup>th</sup> Semester Research) |                                                              |       |            |       |
| CHY014C426                                                                                               | RESEARCH PROJECT – PHASE 2                                   | 500   | 28 + 3/4/5 | 0-0-0 |
| TOTAL CREDIT FOR 2 <sup>nd</sup> YEAR = 52+ 6/8/10                                                       |                                                              |       |            |       |

## SYLLABUS (1<sup>st</sup> SEMESTER)

**Subject Name: Physical Chemistry I**

**Level = 400**

**Subject Code: CHY014C101**

**L-T-P-C: 4-0-0-4**

**Credit Units: 4**

**Scheme of Evaluation: T**

**Objective:** The objective of **Physical Chemistry I** is to understand the concept of statistical thermodynamics & ion-solvent interactions and applications of phase rule.

### Course Outcomes:

| After successful completion of the course, student will be able to |                                                                                                                         |                        |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl. No                                                             | Course Outcome                                                                                                          | Bloom's Taxonomy Level |
| CO1                                                                | <b>Define</b> and gain the knowledge from laws of thermodynamics to solve the complex problems of physical chemistry    | BT1                    |
| CO2                                                                | <b>Explain</b> the concept of thermodynamical parameters and their importance to interpret the spontaneity of reaction. | BT2                    |
| CO3                                                                | <b>Apply</b> the knowledge of chemical kinetics and analyze chemical reactions and reaction mechanism.                  | BT3                    |
| CO4                                                                | <b>Analyze</b> and <b>Explain</b> different energy exchange processes.                                                  | BT4 & BT5              |

### Detailed Syllabus:

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Periods |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <p><b>Classical Thermodynamics</b></p> <p>Thermodynamics of real gases and gas mixtures, Fugacity, relation between fugacity and pressure, variation of fugacity with temperature and pressure, non-ideal solutions, Activity, dependence of activity on temperature and pressure, Fugacity coefficients and activity coefficients–different scales of activity coefficients.</p> <p>Non-equilibrium thermodynamics: Review of basic concepts of force, flow and entropy production, rate of entropy production, entropy production in chemical reactions, coupled forces and flows and phenomenological relations; Onsager reciprocal relations.</p> | 18      |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>II</b>    | <b>Statistical Thermodynamics</b><br><br>Statistical mechanics of systems independent particles- Maxwell Boltzmann distribution law, entropy and probability, calculation of thermodynamic properties for independent particles-molecular partition functions, physical significance of partition function, evaluation of partition function-translational, rotational, vibrational and electronic partition functions. Statistical interpretation of work and heat, thermodynamic properties of ideal monoatomic and diatomic molecules-Suckur-Tetrode equation, calculation of partition functions, thermodynamic function, principles of equipartition. | <b>18</b> |
| <b>III</b>   | <b>Dynamic Electro-chemistry</b><br><br>Ion size factor and ion-solvent interactions–The Born model. Thermodynamic parameters of ion-solvent interactions–structural treatment, the ion-dipole model–its modifications, ion quadrupole and ion–induced dipole interactions.<br>Primary solution–Determination of hydration number, compressibility method and viscosity mobility method.<br>Debye-Hückel theory of ion-ion interactions–derivation, validity and limitations, Debye-Huckel-Onsager treatment and its extension to concentrated solutions.                                                                                                  | <b>18</b> |
| <b>IV</b>    | <b>Phase equilibrium</b><br><br>Phase equilibrium of two-component system, phase diagrams for systems of solid-liquid equilibria involving eutectic, congruent and incongruent melting points, solid solutions.<br>Application of Gibbs phase rule to three component systems – triangular plots–water-acetic acid–chloroform system.<br>Binary solutions: Gibbs-Duhem-Margules equation and its applications to fractional distillation of binary miscible liquids (ideal and non-ideal), azeotropes, partial miscibility of liquids.                                                                                                                     | <b>18</b> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>72</b> |

### **Text Books:**

1. *Physical Chemistry*; Atkins, P.W. and Paula, J. de; 10<sup>th</sup> edition; 2014; Oxford University Press
2. *Principles of Physical Chemistry*; Puri, B.R.; Sharma, L.R.; Pathania, M.S.; 47<sup>th</sup> edition; 2016; Vishal Publishing Company

### **Reference Books:**

1. Glasstone, S.; *Textbook of Physical Chemistry*; 11<sup>th</sup> edition; 2011; VanNostrand company.

- Atkins, P.W. and Paula, J. de; *Elements of Physical Chemistry*; 6<sup>th</sup> edition; 2012; Oxford University Press.
- Kapoor, K. L.; *A textbook of Physical chemistry*; 6<sup>th</sup> edition; 2011; Macmillan, India Ltd.
- Bokris, J.A. and Reddy, A.K.N; *Modern Electrochemistry*; Vols. 1&2; Kluwer Academic Publishers
- Levine, I.; *Physical Chemistry*; 6<sup>th</sup> edition; 2008; McGraw–Hill Science.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

### SYLLABUS (1<sup>st</sup> SEMESTER)

|                                          |                       |                                 |
|------------------------------------------|-----------------------|---------------------------------|
| <b>Subject Name: Organic Chemistry I</b> | <b>Level = 400</b>    | <b>Subject Code: CHY014C102</b> |
| <b>L-T-P-C – 4-0-0-4</b>                 | <b>Credit Units:4</b> | <b>Scheme of Evaluation: T</b>  |

**Objective:** The objective of **Organic Chemistry I** are to provide a thorough knowledge of kinetics of organic reactions, bonding of organic molecules, classifications of reaction with mechanism, stability and reactivity of reaction intermediates, stereochemistry and conformational analysis of molecules.

#### Course Outcomes:

| After successful completion of the course, student will be able to |                                                                                                                           |                        |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                                            | Bloom's Taxonomy Level |
| CO1                                                                | <b>Relate</b> the stereo chemical and conformational aspect of molecules                                                  | BT1                    |
| CO2                                                                | <b>Explain</b> the three dimensional orientation of molecules and its effect on molecular reactivity                      | BT2                    |
| CO3                                                                | <b>Apply</b> the concept of organic chemistry to understand the bond formation and bond breaking of new organic molecule. | BT3                    |
| CO4                                                                | <b>Analyze</b> the reaction mechanism to develop strategy of a new reactions                                              | BT4                    |

**Detailed Syllabus:**

| Modules      | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Periods   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I            | <b>Reaction mechanism</b><br><br>Structure and Reactivity: Types of mechanisms, types of reactions, thermodynamic and kinetic requirements, Hammond postulate, Curtin-Hammett principle, transition states and intermediates, methods of determining mechanisms, isotopic effects. Generation, structure, stability and reactivity of carbocations, carbanions, free radicals, carbenes and nitrenes. Effect of structure on reactivity. The Hammett equation and linear free energy relationship (sigma-rho) relationship, Taft equation.                                                                                      | 12        |
| II           | <b>Stereochemistry and conformational analysis</b><br><br>Acyclic systems upto 4 chiral centres, compounds with asymmetric carbons in branched chain, Klyne-Prelog conformational terminology. Axial chirality, planer chirality. Conformation of cyclic systems: cyclohexene, cyclohexanone, decalin. Optical activity in absence of chiral carbon (biphenyls, allenes and spiranes). Conformational effect on reactivities and physical properties of molecules.                                                                                                                                                              | 12        |
| III          | <b>Bonding in organic compounds</b><br><br>Aromaticity: Concept of Aromaticity, non-aromaticity and antiaromaticity, pseudo aromaticity, homo aromaticity, NMR in aromatic character, Huckel's rule and its limitations, non-benzonoid compounds (aromaticity), alternate and non-alternate hydrocarbons, annulenes, fulvenes, fulvalenes, azulenes, fullerenes.<br>Supramolecular chemistry: Addition compounds: Crown ether complex, Cryptands. Inclusion compounds, Cyclodextrins, Catenanes, Rotaxanes and their applications.                                                                                              | 12        |
| IV           | <b>Disconnection approach</b><br><br>An introduction to synthons and synthetic equivalents, disconnection approach, functional group inter conversion, importance of order of events in organic synthesis, one group C-X and two group C-X disconnections, chemo selectivity, reversal of polarity, cyclisation reactions, amine synthesis. Protecting groups: Principle of protection of alcohol, amine, carbonyl and carboxyl groups. One group C-C disconnection: alcohol, carbonyl compounds, regioselectivity. Use of acetylene in organic synthesis.<br>Diels –Alder reactions, Michael addition and Robinson annulation. | 12        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>48</b> |

**Text books:**

1. *Advanced organic chemistry: Reactions, mechanism and structure*; March Jerry; 7<sup>th</sup> edition; John Wiley; 2015; United States of America.

2. *Advanced organic chemistry*; Carey F. A. and Sundberg R. J.; 5<sup>th</sup> edition; 2007; Plenum.

**Reference books:**

1. Ingold C.K.; *Structure and mechanism in organic chemistry*; 2<sup>nd</sup> edition; Cornell University press.
2. Norman R.O.C. and Coxon J. M.; *Principle of Organic Synthesis*; 3<sup>rd</sup> edition; 1993; Blackie academic professional.
3. Warren S.; *Designing organic synthesis*; 2<sup>nd</sup> edition; 2008; Wiley; UK.
4. Nasipuri D.; *Stereochemistry of organic compounds*; 5<sup>th</sup> edition; 2014; New age international
5. Kalsi P.S.; *Stereochemistry of organic compounds*; 2007; New age international.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

**SYLLABUS (1<sup>st</sup> SEMESTER)**

|                                            |                        |                                 |
|--------------------------------------------|------------------------|---------------------------------|
| <b>Subject Name: Inorganic Chemistry I</b> | <b>Level 400</b>       | <b>Subject Code: CHY014C103</b> |
| <b>L-T-P-C – 4-0-0-4</b>                   | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: T</b>  |

**Objective:** The objective of **Inorganic Chemistry I** is to provide detailed knowledge on the nature of interactions in compounds and in-depth information about acidic/basic and redox properties of inorganic materials.

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                                                                 |                        |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                                                                  | Bloom's Taxonomy Level |
| CO1                                                                | <b>Relate</b> the structure, bonding and properties of coordination and organometallic compounds.                                               | BT1                    |
| CO2                                                                | <b>Compare</b> the interaction between ligands and central atom to solve the problems related to their structure, stability and reactivity      | BT2                    |
| CO3                                                                | <b>Apply</b> and <b>analyse</b> the knowledge of acidic/basic and redox nature of the inorganic compounds to predict the nature of the reaction | BT 3 & 4               |
| CO4                                                                | <b>Evaluate</b> and <b>construct</b> new inorganic compounds to varieties of transition and non-transition elements                             | BT 5 & 6               |

**Detailed Syllabus:**

| Modules      | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Periods   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I            | <b>Chemical Bonding</b><br><br>Chemical bonding of simple inorganic covalent compounds-molecular orbital treatments, hybridization, understanding molecular properties from bonding, molecular orbital theory of homo and heteronuclear diatomic, molecular orbitals of polyatomic molecules, molecular shape in terms of molecular orbitals – Walsh diagrams, atomic and ionic radii, bond length, bond strength, van der Waals forces, effect of hydrogen bonding and other chemical forces on melting and boiling points and solubility. | 12        |
| II           | <b>Coordination Chemistry I</b><br><br>Crystal field theory, ligand field theory, splitting of d-orbitals, crystal field stabilization energies in weak field and strong field, octahedral site preference energy, tetragonal distortion and Jahn-Teller effect, lattice energy, hydration enthalpy and stability of complexes (Irving-Williams order).                                                                                                                                                                                     | 12        |
| III          | <b>Acid Base and Redox Chemistry</b><br><br>Hard and soft acid-base concept, strength of oxo acids and halo acids, strength of inorganic bases, periodic trends in acidity and basicity of hydrides, oxides, oxyacids of non-transition elements.<br>Standard electrode potentials, pH dependence of electrode potentials, redox stability of metal ions in water, Latimer and Frost diagrams.                                                                                                                                              | 12        |
| IV           | <b>Non-Transition Metal Chemistry</b><br><br>Synthesis, properties, structure and bonding of nitrogen, phosphorous, sulfur, pseudohalogen, interhalogen and xenon compounds, boranes, carboranes, metallocarboranes, borazines, phosphazenes, sulfur-nitrogen compounds, silicates, silicones.                                                                                                                                                                                                                                              | 12        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>48</b> |

**Text Books:**

1. *Concise Inorganic Chemistry*; Lee, J.D.; 5<sup>th</sup> edition; 2013; John Wiley and Sons Ltd.; Indian Edition.
2. *Inorganic Chemistry*; Atkins, P., Overton, T., Rourke, J., Weller, M. and Armstrong, F.; 6<sup>th</sup> edition; 2014; Oxford University Press; Indian edition.
3. *Inorganic Chemistry Principles of Structure and Reactivity*; Huheey, J.E, Keiter, E. A., Keiter, R. L and Medhi, O. K.; 4<sup>th</sup> edition; 2007; Pearson Education.

**Reference Books:**

1. Wells, A.F.; *Structural Inorganic Chemistry*; 3<sup>rd</sup> edition; 2012; Oxford Science Publishers.
2. Cotton, F.A., Wilkinson, G., Murillo, A., Bochmann M.; *Advanced Inorganic Chemistry*; 6<sup>th</sup> edition; 1999; Wiley Interscience; New York.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

### SYLLABUS (1<sup>st</sup> SEMESTER)

|                                        |                        |                                 |
|----------------------------------------|------------------------|---------------------------------|
| <b>Subject Name: Quantum Chemistry</b> | <b>Level = 400</b>     | <b>Subject Code: CHY014C104</b> |
| <b>L-T-P: 4-0-0-4</b>                  | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: T</b>  |

**Objective:** The objectives of **Quantum Chemistry** are to provide a thorough knowledge of quantum mechanics, solve and apply the approximate methods in real molecules.

#### Course Outcomes:

| After successful completion of the course, the students will be able to |                                                                                          |                        |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                                   | Course Outcome                                                                           | Bloom's Taxonomy Level |
| CO1                                                                     | Define the postulates and interpret the basic principles of quantum mechanics            | BT1 & BT2              |
| CO2                                                                     | Apply quantum mechanical treatment to various models                                     | BT3                    |
| CO3                                                                     | Analyze and Explain the approximation methods in terms of real molecules                 | BT4 & BT5              |
| CO4                                                                     | Construct the various orbital theories to solve for energy values of different molecules | BT6                    |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Periods   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I       | <b>Basic Principles of Quantum Mechanics</b><br><br>Origin of the quantum theory, wave functions of one-particle and many-particles system, probability density, well-behaved functions, normal and orthogonal functions.<br>Operators in quantum mechanics, eigen values and eigen functions, Hermitian operators and their properties, commutation of operators, postulates of quantum mechanics, expectation values of observable properties.<br>Angular momentum of a one-particle system, spin and orbital angular momentum. | 18        |
| II      | <b>Some Exactly Solvable Problems in Quantum Mechanics</b><br><br>Quantum mechanical treatment of translational motion of a particle, particle in one- and three-dimensional boxes, concept of degeneracy, harmonic oscillator, rotational motion of a particle: particle on a ring, rigid rotator, hydrogen and hydrogen like atoms, graphical presentation of orbitals (s, p and d), radial and angular probability distribution plots.                                                                                         | 18        |
| III     | <b>Approximate Methods</b><br><br>Need for approximation methods, perturbation and variation methods and their application to Helium atom, symmetric and antisymmetric wave functions, Pauli's exclusion principles, many electron atoms, Slater determinants, qualitative treatment of Hartree theory and Hartree-Fock SCF procedure.                                                                                                                                                                                            | 18        |
| IV      | <b>Chemical Bonding</b><br><br>Born-Oppenheimer approximation, separation of electronic and nuclear motion, hydrogen molecule ion: linear combination of atomic orbital (LCAO)-molecular (MO) theory, valence bond (VB) and MO (LCAO) treatment of hydrogen molecule, comparison of MO and VB treatments and their equivalence limit, Huckel MO theory, FMO.                                                                                                                                                                      | 18        |
|         | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>72</b> |

**Text Books:**

1. *Quantum Chemistry*; Levine, I.N.; 5<sup>th</sup> edition, 2000; Prentice Hall of India
2. *Quantum Chemistry*; Prasad, R.K.; 4<sup>th</sup> edition; 2009; New Age International Publishers Limited

**Reference Books:**

- Chandra, A .K.; *Introductory Quantum Chemistry*; 4<sup>th</sup> edition; 2006;TataMcGrawHill
- Sen, B.K.; *Quantum Chemistry Including Spectroscopy*; 4<sup>th</sup> edition; 2011; Kalyani Publishers, New Delhi
- McQuarrie, D.A.; *Quantum Chemistry*; 2<sup>nd</sup> edition; 2011; Viva Books Pvt Ltd
- Atkins, P.W and S.F. Ronald; *Molecular Quantum Mechanics*; 5<sup>th</sup> edition; Oxford University Press

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

**SYLLABUS (1<sup>st</sup> SEMESTER)****Subject Name: Inorganic Chemistry Lab****Subject Code: CHY014C115****L-T-P-C – 0-0-8-4****Credit Units: 4****Scheme of Evaluation: (P)**

**Objective:** The objective of **Inorganic Chemistry Lab** is to provide hands on training to synthesize inorganic metal complex and analyze quantitatively the acid/base and redox reactions through titrimetric methods

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                                           |                        |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------|
| SI No                                                              | Course Outcome                                                                                                            | Bloom's Taxonomy Level |
| CO1                                                                | <b>Relate</b> knowledge of inorganic chemistry in scientific and technological aspects and synthesise inorganic complexes | BT1                    |
| CO2                                                                | <b>Compare</b> the interaction between ligands and central atom to predict their structure, stability and reactivity      | BT2                    |
| CO3                                                                | <b>Apply</b> and <b>analyze</b> the knowledge of acidic/basic and redox titrations in quantitative analysis               | BT3 & BT4              |
| CO4                                                                | <b>Evaluate</b> and <b>construct</b> new green methods for synthesis of inorganic compounds.                              | BT5 & BT6              |

**Detailed Syllabus:****List of Experiments:**

1) Preparation of following inorganic compounds:

- Potassium trioxalatoaluminate (III) trihydrate,  $K_3[Al(C_2O_4)_3].3H_2O$
- Hexaamminenickel(II) chloride,  $[Ni(NH_3)_6]Cl_2$
- Potassiumtrioxalatoferrate(III) trihydrate,  $K_3[Fe(C_2O_4)_3].3H_2O$

2) Preparation of potassium trioxalatochromate(III) trihydrate  $K_3[Cr(C_2O_4)_3].3H_2O$  and determination of concentration of chromium and oxalate ion.

- 3) Quantitative analysis of ore/alloy:
  - i. Determination of the amount of calcium in the limestone sample
  - ii. Determination of the percentage of copper in Brass sample
  - iii. Determination of the percentage of iron in steel sample
- 4) Determination of concentration of components in a mixture
  - i. Estimation of  $\text{Fe}^{\text{II}}$  and  $\text{Fe}^{\text{III}}$  in a mixture
  - ii. Estimation of  $\text{Na}_2\text{CO}_3$  and  $\text{NaHCO}_3$  in amixture
- 5) Estimation of alkali content of antacid tablets.
- 6) Green Chemistry experiments:
  - i. Recovery and reuse of sulfur dioxide (Obendrauf's Method)
  - ii. Green synthesis of Tetrabutylammonium tribromide(TBATB)
  - iii. Preparation of Bis(acetylacetonato)copper(II)

### **Text Books:**

1. *Green Chemistry Experiments: A Monograph*; Sharma R.K., Sidhwani I.T., Choudhuri M.K.; 1<sup>st</sup> edition (December, 2012); I K International Publishing House.
2. *Vogel's Qualitative Inorganic Analysis*; Svehla G. and Sivasankar B; 7<sup>th</sup> edition; Pearson.

### **Reference Books:**

1. Barua, S.; *A text Book of Practical Chemistry*; 2<sup>th</sup> edition; 2016; Kalyani Publishers.
2. Mendham J., Denney R.C., Barnes J.D. and. Thomas M.J.K.; *Vogel's Textbook of Quantitative Chemical Analysis*, 6<sup>th</sup> edition, 3<sup>rd</sup> Indian Reprint, 2003, Pearson Education Pvt. Ltd., New Delhi

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

## SYLLABUS (2<sup>nd</sup> SEMESTER)

**Subject Name: Physical Chemistry II**

**Level:400**

**Subject Code: CHY014C201**

**L-T-P: 4-0-0-4**

**CreditUnits:4**

**Scheme of Evaluation: T**

**Objective:** The objectives of **Physical Chemistry II** are to study the reaction mechanism, theories of reaction rate, kinetic and thermodynamic control of reactions and to understand the properties of polymers and techniques involved in polymerization.

**Course Outcomes:**

| After successful completion of the course, the students will be able to |                                                                                        |                        |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------|
| S. No.                                                                  | Course Outcome                                                                         | Bloom's Taxonomy Level |
| CO1                                                                     | Define rate laws and <b>understand</b> the theories that govern the reaction rates.    | BT1 & BT2              |
| CO2                                                                     | <b>Build</b> mechanisms of catalysis and <b>apply</b> catalytic process in industries. | BT3 & BT4              |
| CO3                                                                     | <b>Determine</b> the degree of polymerization and molecular weights of polymers        | BT5                    |
| CO4                                                                     | <b>Construct</b> and <b>compare</b> various adsorption isotherms                       | BT5 & BT6              |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Periods |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <p><b>Reaction Kinetics</b></p> <p>Empirical rate law and temperature dependence, steady-state approximation, determination of reaction mechanisms, oscillating reactions: Belousov-Zhabotinski reaction, chain reaction: alkane pyrolysis, branched chain reactions: the hydrogen oxygen reaction, concept of explosion limit.</p> <p>Collision theory, estimation of activation energy and the calculation of potential energy surface, transition state theory (TST) of bimolecular reactions, Eyring equation, kinetic and thermodynamic control of reactions, Lindemann-Hinshelwood theory of unimolecular reactions.</p>                              | 18      |
| II      | <p><b>Catalysis</b></p> <p>Types of catalyst, specificity and selectivity, effect of particle size and efficiency of nanoparticles as catalysts.</p> <p><i>Homogeneous catalysts:</i> Mechanism of homogeneous catalysis, acid-base catalysis, enzyme catalyzed reactions, Michaelis-Menten mechanism, effect of pH and temperature, enzyme inhibitor, role of transition metal ions as catalyst with special reference to Cu, Pd, Pt, Co, Ru and Rh, acid-base catalysis.</p> <p><i>Heterogeneous catalysts:</i> Kinetics of heterogeneous catalysis: Langmuir-Hinshelwood model, clays, zeolites and their use as catalysts in cracking of petroleum.</p> | 18      |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| III | <b>Adsorption and Aggregation</b><br><br>Adsorption of gases on solid surfaces: Langmuir's theory and its limitations, derivation of BET equation: determination of surface area of an adsorbent, adsorption in liquid systems: Gibbs adsorption isotherm.<br><br>Colloidal system, optical, kinetic and electrical properties of colloids, electrophoresis, electro-osmosis, size determination of colloidal particles, coagulation of colloidal solutions.<br><br>Surface active agents and their classifications, aggregation /micellization of surfactants, hydrophobic interaction, critical micellar concentration (CMC), factors affecting the CMC of surfactants, thermodynamics of micellization: phase separation and mass action models, microemulsion, reverse micelles. | 18        |
| IV  | <b>Macromolecules</b><br>Polymers and degree of polymerization, classification of polymers, mechanism of polymerization, concept of number average and mass average molecular weight of a polymer, methods of determining molecular weights (osmometry, viscometry, light scattering and sedimentation equilibrium methods), chain configuration of macromolecules: root mean square end to end distance and radius of gyration.                                                                                                                                                                                                                                                                                                                                                     | 18        |
|     | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>72</b> |

#### **Text Books:**

1. *Physical Chemistry*; Atkins, P.W. and Paula, J. de; 10<sup>th</sup> edition; 2014; Oxford University Press
2. *Principles of Physical Chemistry*; Puri, B.R.; Sharma, L.R.; Pathania, M.S.; 47<sup>th</sup> edition; 2016; Vishal Publishing Company

#### **Reference Books:**

1. Glasstone, S.; *Text book of Physical Chemistry*; 11<sup>th</sup> edition; 2011; Van Nostrand company.
2. Atkins, P.W. and Paula, J. de; *Elements of Physical Chemistry*; 6<sup>th</sup> edition, 2012; Oxford University Press.
3. Kapoor, K. L.; *A textbook of Physical chemistry*; 6<sup>th</sup> edition; 2011; Macmillan, India Ltd.
4. Levine, I.; *Physical Chemistry*; 6<sup>th</sup> edition; 2008; McGraw–Hill Science
5. Billmeyer, F.W.; *Text Book of Polymer Science*; 2<sup>nd</sup> edition; 1971, John.Wiley, London
6. Gowariker, V.R.; Viswanathan, N.V.; Sreedhar, T.; *Polymer Science*; 1<sup>st</sup> edition; 1986; Wiley Eastern, New Delhi
7. Mishra, G.S.; *Introductory Polymer Chemistry*; 5<sup>th</sup> edition; 2007; New Age International  
Gates, B.C.; *Catalytic Chemistry*; 1<sup>st</sup> edition; 1992; John Wiley & Sons, New York

| <b>Credit Distribution</b> |                  |                              |
|----------------------------|------------------|------------------------------|
| <b>Theory/ Tutorial</b>    | <b>Practicum</b> | <b>Experimental Learning</b> |
| <b>72</b>                  | <b>---</b>       | <b>48</b>                    |

## SYLLABUS (2<sup>nd</sup> SEMESTER)

|                                           |                        |                                  |
|-------------------------------------------|------------------------|----------------------------------|
| <b>Subject Name: Organic Chemistry II</b> | <b>Level: 400</b>      | <b>Subject Code: CHY014C202</b>  |
| <b>L-T-P-C – 4-0-0-4</b>                  | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: (T)</b> |

**Objective:** To provide the synthetic methodology of organic compounds. To cover the reagents used in organic synthesis. To throw some lights on rearrangement reactions. To discuss about Organometallic reactions.

### Course Outcomes:

| After successful completion of the course, student will be able to |                                                                                                                                                                                  |                        |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                                                                                                   | Bloom's Taxonomy Level |
| CO1                                                                | <b>Understand</b> the chemistry of reagents and name reactions and their synthetic applications.                                                                                 | BT1                    |
| CO2                                                                | <b>Explain</b> the mechanisms of various rearrangement reactions                                                                                                                 | BT2                    |
| CO3                                                                | <b>Apply</b> the concept of reagents, ylides and organometallic compounds for the conversion of one functional group into other functional group in one or more number of steps. | BT3                    |
| CO4                                                                | <b>Analyze</b> the structures knowledge of commercially important molecules.                                                                                                     | BT4                    |

### Detailed Syllabus:

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Periods |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <b>Reagents in organic synthesis</b><br><br>Complex metal hydrides, DIBAL-H, Gilman's reagent, LDA, DCC, 1,3-propane dithiane, Trimethyl-silyl-tin hydride, Tri-n-butyl-tin hydride, Woodwards and Prevost hydroxylation, DDQ, SeO <sub>2</sub> , PPC, PDC, Merifield resins, Peterson's synthesis, Baker's yeast, Chromic acid, Potassium dichromate, Jones reagent, Collins reagent, Birch reduction, Periodic acid, Lead tetra acetate, Osmium tetra oxide, Ozonolysis, m-CPBA, Wittig reagent                               | 12      |
| II      | <b>Selective Name Reactions</b><br><br>Aldol, Perkin, Stobbe, Dieckmann condensation, Diels-Alder reactions, Robinson annulation, Michael, Mannich, Stork enamine, Sharpless asymmetric epoxidation, Barton, Ene, Hoffman-Löffler-Freytag, Shapiro, Chichibabin, Cannizzaro, Bayer-Hilman, Darzens, Benzoin condensation, Knoevenagel, Reimer-Tieman reaction, Wolf-Kishner reduction, Clemmenson reduction, Moningo reduction, Meerwein-Pondorf-Verley reduction, Oppenauer oxidation, Dess-Martin oxidation, Swern oxidation. | 12      |

|              |                                                                                                                                                                                                                                                                                                                                |           |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>III</b>   | <b>Rearrangement reactions</b><br><br>Wagner-Meerwein, Pinacol-pinacolone, Wolff, Arndt-Eistert synthesis, Hofmann, Curtius, Schmidt, Lossen, Beckmann, Bayer-Villiger, Favorski, Benzillic acid rearrangement, Stevens, Wittig, Claisen, Cope.                                                                                | <b>12</b> |
| <b>IV</b>    | <b>Ylides and organometallic chemistry:</b><br><br>Methods of generation, properties and reactions of organo magnesium, lithium, cadmium, zinc, copper, boron. Grignard reagent and its application, Reformatsky reaction and its application. Phosphorous and sulfur ylides: methods of generation, properties and reactions. | <b>12</b> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                | <b>48</b> |

### **Text books:**

1. *Advanced organic chemistry: Reactions, mechanism and structure*; March Jerry; 7<sup>th</sup> edition; John Wiley.
2. *Advanced organic chemistry*; Carey F. A. and Sundberg R. J.; 5<sup>th</sup> edition; Plenum.
3. *Principle of Organic Synthesis*; Norman R.O.C. and Coxon J. M.; 3<sup>rd</sup> edition; Blackie academic professional.
4. *A guide book to mechanism in organic chemistry*; Sykes Peter; 6<sup>th</sup> edition; 2013, Longman.

### **Reference books:**

1. Carruthers W., *Some modern methods of organic synthesis*; 4<sup>th</sup> edition; Cambridge University press.
2. Clayden J., Greeves N. and Warren S., *Organic chemistry*; 2<sup>nd</sup> edition; Oxford University press.

| <b>Credit Distribution</b> |                  |                              |
|----------------------------|------------------|------------------------------|
| <b>Theory/ Tutorial</b>    | <b>Practicum</b> | <b>Experimental Learning</b> |
| <b>72</b>                  | <b>---</b>       | <b>48</b>                    |

| <b>SYLLABUS (2<sup>nd</sup> SEMESTER)</b>   |                       |                                  |
|---------------------------------------------|-----------------------|----------------------------------|
| <b>Subject Name: Inorganic Chemistry II</b> | <b>Level: 400</b>     | <b>Subject Code: CHY014C203</b>  |
| <b>L-T-P-C-4-0-0-4</b>                      | <b>Credit Units:4</b> | <b>Scheme of Evaluation: (T)</b> |

**Objective:** The objective of **Inorganic Chemistry-II** is to understand and apply the concepts of Inorganic reaction mechanism, inorganic spectroscopy and organometallic compounds. It will also provide in-depth knowledge related to the concepts of symmetry and structure.

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                                                                 |                        |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                                                                  | Bloom's Taxonomy Level |
| CO1                                                                | <b>Describe</b> the mechanism of inorganic compounds                                                                                            | BT1                    |
| CO2                                                                | <b>Interpret</b> general features of the electronic absorption spectra including the charge transfer spectra of the transition metal complexes. | BT2                    |
| CO3                                                                | <b>Make use</b> of the 18-electron rule to predict the stability and chemistry of organometallic and transition metal complexes.                | BT3                    |
| CO 4                                                               | <b>Analyse</b> structure and symmetry of inorganic compounds                                                                                    | BT4                    |

**Detailed Syllabus:**

| Modules      | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Periods   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I            | <b>Inorganic Reaction Mechanism</b><br><br>Substitution in octahedral and square planar complexes, lability, trans-effect, conjugate base mechanism, racemisation, electron transfer reactions: inner sphere and outer sphere mechanism, Marcus theory, inorganic photochemistry: Photo substitution and photo redox reactions of chromium, cobalt and ruthenium compounds, Adamson's rules.                                                                                                                                                                                                              | 10        |
| II           | <b>Coordination Chemistry II</b><br><br>Interpretation of the general features of the electronic absorption spectra including the charge transfer spectra of the transition metal complexes in aqueous solutions, spin-orbit coupling constant and interelectronic repulsion parameters in complex ion terms-vs-free ion terms, vibronic coupling, intensity stealing, band broadening, spectrochemical series, nephelauxetic series, structural distortion and lowering of symmetry, electronic, steric effect on energy levels, magnetic properties, quenching of orbital moment and spin only formula. | 10        |
| III          | <b>Organometallics</b><br><br>18 electron rule, metal carbonyls, nitrosyls, carbonyl hydrides, isolobal analogy, dioxygen and dinitrogen compounds, metal alkyls, carbenes, carbynes, alkenes, alkynes, and allyl complexes, hydrides, metallocenes, metal arene complexes, carbonylate anions, oxidative addition and reductive elimination, insertion and elimination reactions, homogeneous and heterogeneous catalysis, fluxional molecules.                                                                                                                                                          | 10        |
| IV           | <b>Symmetry and Structure</b><br><br>Symmetry elements and operations, equivalent symmetry elements and equivalent atoms, symmetry point groups with examples from inorganic compounds, groups of very high symmetry, molecular dissymmetry and optical activity, systematic procedure for symmetry classification of molecules and illustrative examples, molecular symmetry for compounds having co-ordination numbers 2 to 9.                                                                                                                                                                          | 10        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>40</b> |

**Text Books:**

1. *Concise Inorganic Chemistry*; Lee, J.D.; 5<sup>th</sup> edition; 2013; John Wiley and Sons Ltd.; Indian Edition.
2. *Inorganic Chemistry*; Atkins, P., Overton, T., Rourke, J., Weller, M. and Armstrong, F.; 6<sup>th</sup> edition; 2014; Oxford University Press; Indian edition.
3. *Inorganic Chemistry Principles of Structure and Reactivity*; Huheey, J.E, Keiter, E. A., Keiter, R. L and Medhi, O. K.; 4<sup>th</sup> edition; 2007; Pearson Education.

**Reference Books:**

1. Wells, A.F.; *Structural Inorganic Chemistry*; 3<sup>rd</sup> edition; 2012; Oxford Science Publishers.
2. Cotton, F.A., Wilkinson, G., Murillo, A., Bochmann M.; *Advanced Inorganic Chemistry*; 6<sup>th</sup> edition; 1999; Wiley Interscience; New York.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

**SYLLABUS (2<sup>nd</sup> SEMESTER)**

|                                           |                        |                                  |
|-------------------------------------------|------------------------|----------------------------------|
| <b>Subject Name: Analytical Chemistry</b> | <b>Level: 400</b>      | <b>Subject Code: CHY014C204</b>  |
| <b>L-T-P-C-4-0-0-4</b>                    | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: (T)</b> |

**Objective:** The objective of **Analytical Chemistry** is to provide a thorough background of principles which are important to chemical analysis and to develop the skills in the proper handling of apparatus and recording of data.

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                        |                        |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------|
| SI No                                                              | Course Outcome                                                                                         | Bloom's Taxonomy Level |
| CO1                                                                | <b>Relate</b> the importance of chemical analysis with instrumental techniques                         | BT 1                   |
| CO2                                                                | <b>Explain</b> the concept of purification, separation and identification of the analytes.             | BT 2                   |
| CO3                                                                | <b>Apply and predict</b> the nature of the various samples through optical and electrothermal analysis | BT3 & BT6              |
| CO4                                                                | <b>Analyze and measure</b> the accuracy and errors in experimental data.                               | BT4 & BT5              |

**Detailed Syllabus:**

| <b>Modules</b> | <b>Topics &amp; Course content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Periods</b> |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>I</b>       | <b>Treatment of Analytical Data</b><br><br>Definition and brief idea of the following terms: Significant figures, accuracy and precision, mean, median, variance, confidence limits, deviation, relative mean deviation, standard deviation.<br>Types of errors in chemical analysis: Determinate and indeterminate error, absolute errors, relative errors, constant and proportional errors, minimization of determinates errors.                                                                                                                                                                                                                           | <b>12</b>      |
| <b>II</b>      | <b>Chromatography and Thermal Methods of Gravimetry</b><br><br>Theory of chromatography, retention time, classification of chromatography, chromatographic techniques – principles, experimental techniques and applications of Gas Chromatography, Liquid Chromatography, Column Chromatography, Thin Layer Chromatography, High-Performance Liquid Chromatography.<br>Thermal Methods: Principle and application of thermal methods of analysis — TGA, DTA and DSC.                                                                                                                                                                                         | <b>12</b>      |
| <b>III</b>     | <b>Optical Methods</b><br><br>Fundamental laws of spectrophotometry, nephelometry, turbidometry and fluorimetry. Spectrophotometric titrations.<br>Atomic emission spectrometry: Excitation sources (flame, AC and DC arc), spark, inductively coupled plasma, glow discharge, laser microprobes, flame structure, instrumentation and qualitative and quantitative analysis. Atomic absorption spectrometry: Sample atomization techniques, instrumentation, interferences, background correction, and analytical applications.<br>Theory, instrumentation and applications of: Atomic fluorescence spectrometry, photoelectron spectroscopy, SEM, TEM, AFM. | <b>12</b>      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| IV | <b>Electro Analytical Methods</b><br><br>Potentiometry: Techniques based on potential measurements, direct potentiometric systems, different types of indicator electrodes, limitations of glass electrode, applications in pH measurements, modern modifications. Polarography: Micro electrode and their specialities, potential and current variations at the micro electrode systems, conventional techniques for concentration determination, limitations of detection at lower concentrations, techniques of improving detection limit-rapid scan, ac, pulse, differential pulse square wave polarographic techniques. Applications of polarography. Amperometry: Biamperometry, amperometric titrations. Coulometry: Primary and secondary coulometry, advantages of coulometric titrations, applications. Principle of chronopotentiometry. Anodic stripping voltammetry: Different types of electrodes and improvements of lower detection limits. Voltammetric sensors. | 12        |
|    | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>48</b> |

#### Text books:

1. *Vogel's Qualitative Inorganic Analysis*; Svehla G. and Sivasankar B.; 7<sup>th</sup> edition; Pearson.
2. *Fundamental of Analytical Chemistry*; Skoog D.M.; Holler and Crouch, West, VIII Edition, 2005, Saunders College Publishing, New York.

#### Reference Books:

1. Mendham J., Denney R.C., Barnes J.D. and Thomas M.J.K.; *Vogel's Textbook of Quantitative Chemical Analysis*, 6<sup>th</sup> edition, 3<sup>rd</sup> Indian Reprint, 2003, Pearson Education Pvt. Ltd., New Delhi.
2. Day R.A. and A.L., *Quantitative Analysis*; Underwood, 6<sup>th</sup> edition, 1993 Prentice Hall, Inc. New Delhi.
3. *Bioinorganic Chemistry*, 3<sup>rd</sup> edition; 2006; Wiley.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

## SYLLABUS (2<sup>nd</sup> SEMESTER)

|                                            |                        |                                  |
|--------------------------------------------|------------------------|----------------------------------|
| <b>Subject Name: Organic Chemistry Lab</b> | <b>Level:400</b>       | <b>Subject Code: CHY014C215</b>  |
| <b>L-T-P-C – 0-0-8-4</b>                   | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: (P)</b> |

**Objective:** The objectives of **Organic Chemistry Laboratory** is to provide knowledge of organic chemistry in scientific and technological aspects, to develop curiosity and interest as well as to improve the understanding of the concepts and application of organic chemistry

### Course Outcomes:

| After successful completion of the course, student will be able to |                                                                                    |                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                     | Bloom's Taxonomy Level |
| CO1                                                                | <b>Recall</b> the chemistry behind qualitative analysis of organic compounds       | BT1                    |
| CO2                                                                | <b>Classify</b> compounds depending upon the functional groups.                    | BT2                    |
| CO3                                                                | <b>Analyze</b> and <b>Apply</b> the knowledge for synthesis of organic derivative. | BT3 & BT4              |
| CO4                                                                | <b>Measure</b> and <b>solve</b> the compounds with independent thinking.           | BT5 & BT6              |

### Detailed Syllabus:

#### **List of Experiments:**

#### **1) Qualitative analysis of binary mixtures of organic compounds (min. four samples)**

- Detection of special elements (N, Cl, S)
- Solubility and Classification
- Detection of the functional groups by systematic chemical tests
- Preparation of derivative of each functional group, purification of crude product by crystallization.
- Determination of melting point of the given samples and derivatives

#### **2) Chromatography experiments: (any one)**

- TLC separation and identification.
- Separation techniques of organic compounds by column chromatography and their spectroscopic identification

#### **3) Experiments on Natural products: (any one)**

- Determination of saponification equivalent of ester/ Saponification of Vegetable Oil, biodiesel.
- Extraction of carotenoids/Lycopene from a natural source.

#### 4) Synthesis of organic compounds common reagents:

##### Two steps synthesis- (any one)

(a) Benzoin – Benzil – Benzilic acid) : Base catalysed, rearrangement.

##### Three steps synthesis- (any one)

(a) Oxidation reaction (b) Reduction reaction (c) Nucleophilic substitution; (d) Cycloaddition reaction; (e) Condensation reaction; (f) Aromatic electrophilic substitution; (g) Preparation of dyes, (h) Heterocyclic synthesis, etc. (e.g., Synthesis of antibacterial compound such as Sulphanilamide, synthesis of p-Amino Benzoic Acid) Preparation of p-acetotoluidine to toluidine

#### 5. Quantitative analysis: (any two)

- (a) Determination of equivalent mass of an acid by direct titration method
- (b) Estimation of alcohol content in a sample using UV-visible spectrometer
- (c) Estimation of sugars using titrimetric (redox) methods.
- (d) Glucose and sucrose in a mixture.

#### 6) Green experiments: (any one)

- (i) Microwave assisted synthesis of some organic compounds
- (ii) Coenzyme catalysed benzoin condensation (thiamine hydrochloride catalysed synthesis of benzoin)
- (iii) Nitration of phenol using calcium nitrate tetrahydrate, acetic acid and salicylic acid
- (iv) Acetylation of primary amine (Preparation of acetanilide) using aniline, acetic acid and zinc dust.

#### Text books:

1. *Vogel's Textbook of Practical Organic Chemistry*, Vogel A.I., Aurther I., 5<sup>th</sup> Edition, 2005, Pearson.
2. *Advanced Practical Organic Chemistry*, Agarwal O. P., 2<sup>nd</sup> Edition, 2014, Goel Publishing.
3. *Green Chemistry Experiments: A Monograph*; Sharma R.K., Sidhwani I.T., Choudhuri M.K.; 1<sup>st</sup> edition (December, 2012); I K International Publishing House.

### SYLLABUS (3<sup>rd</sup> SEMESTER)

Subject Name: Spectroscopy-I

Level: 500

Subject Code: CHY014C301

L-T-P-C-4-0-0-4

Credit Units: 4

Scheme of Evaluation: (T)

**Objective:** The objectives of **Spectroscopy-I** is to provide the knowledge, techniques and principles of spectroscopy and to develop the ability to solve problems.

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                          |                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                           | Bloom's Taxonomy Level |
| CO1                                                                | <b>Understand</b> the chemistry of molecular spectroscopy                                | BT1                    |
| CO2                                                                | <b>Demonstrate</b> the structural characteristics of various types of compounds.         | BT2                    |
| CO3                                                                | <b>Analyze</b> and <b>Apply</b> the knowledge of spectroscopy to practical applications. | BT3 & BT4              |
| CO4                                                                | <b>Measure</b> and <b>solve</b> the synthesized organic compounds.                       | BT5 & BT6              |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Periods |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <b>Basic theory of spectroscopy</b><br>Electromagnetic spectrum, interaction of electromagnetic radiation with molecular systems.<br>Spectroscopic transition- absorption, emission, reflection, polarization and scattering processes.<br>Natural line width and broadening- intensity of spectral transitions, selection rules; sampling techniques in different branches of spectroscopy.<br>Electronic transitions, the Frank-Condon principle, ground and first excited states of diatomic molecules, selection rules on the basis of the symmetry properties of the electronic states; vibronic transitions.                                                                                                          | 12      |
| II      | <b>Basic theory of UV-Visible and IR spectroscopy with its application</b><br>UV-Visible spectroscopy: Basic principle, process of electronic excitation n-p and p-p transitions, transition probability, solvent effect, factors affecting position and intensity of absorption bands, spectra of dienes, polyenes and unsaturated ketones, calculation of $\lambda_{\text{max}}$ , Woodward –Fieser rules.<br>IR Spectroscopy: Stretching vibrations, Hooke's Law, stretching and bending vibrations.<br>Application of IR spectroscopy: Identity of samples, effects of substitution, conjugation, bond angle, and hydrogen bonding on vibrational frequencies. Detection of inter-and intra-molecular hydrogen bonding. | 12      |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| III | <p><b>Basic theory of <math>^1\text{H}</math> and <math>^{13}\text{C}</math> NMR spectroscopy and its application</b></p> <p>Nuclear Magnetic Spectroscopy: Basic instrumentation, nuclear spin, nuclear resonance, chemical shift and its measurements, shielding and deshielding, spin-spin interaction, coupling constant, Karplus equation.</p> <p>Simplification of spectra by use of Lanthanide shift-reagents and high magnetic fields. Deuterium exchange technique in the determination of labile hydrogen, spin decoupling, and Nuclear Overhauser effect (NOE).</p> <p>Two-dimensional NMR spectroscopy: COSY, NOESY, DEPT.</p> <p><math>^{13}\text{C}</math> NMR spectra: Basic theory of <math>^{13}\text{C}</math> NMR spectroscopy.</p> <p>Application of <math>^1\text{H}</math> and <math>^{13}\text{C}</math> NMR spectroscopy in the structure elucidation of simple molecules.</p> | 12        |
| IV  | <p><b>Basic theory of Mass spectroscopy and its application</b></p> <p>Mass Spectroscopy: Basic instrumentation, molecular ion peak, ion production-EI, CI, MALDI techniques. Mass spectral fragmentation of typical organic compounds, common functional groups, McLafferty rearrangement.</p> <p>Application of Mass spectroscopy, examples of mass spectral Fragmentation of organic compounds with respect to their structure determination.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 12        |
|     | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>48</b> |

**Textbooks:**

1. *Organic spectroscopy*; Kemp W.; 3<sup>rd</sup> edition; 1993; ELBS with Mcmillan.

**Reference books:**

1. Dyer J. R.; *Application of spectroscopy in organic compounds*; 1994; Prentice Hall; New Delhi.
2. Silverstein Robert M., Webster Francis X., Kiemle David J.; *Spectrometric identification of organic compounds*; 8<sup>th</sup> edition; 2017; Wiley

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

**Subject Name: Biochemistry and Bioinorganic Chemistry Level: 500 Subject Code: CHY014C302**

**L-T-P-C-4-0-0-4**

**Credit Units: 4**

**Scheme of Evaluation: (T)**

**Objective:** The objectives of **Biochemistry and Bioinorganic Chemistry** is to provide the knowledge of molecular structure and interactions present in various biomolecules and bioinorganic compounds that help in understanding functioning and organizing of living organisms.

**Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                                      |                        |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                                       | Bloom's Taxonomy Level |
| CO1                                                                | <b>Relate</b> importance of various Essential elements and metals in biological system.                              | BT1                    |
| CO2                                                                | <b>Explain</b> the structure and biological functions of metalloenzymes.                                             | BT2                    |
| CO3                                                                | <b>Apply</b> the knowledge of effect of Biology-Chemistry interaction on living organism                             | BT3                    |
| CO4                                                                | <b>Analyze</b> the role of iron in biological system and various chemical reactions happening in biological systems. | BT4                    |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Periods |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <b>Essential elements and metals in biological system</b><br><br>Essential and trace elements in the biological systems, metals in life, basic reactions in the biological systems and the roles of metal ions in biological processes. Ion transport (active) across biological membrane and its significance, mechanism of Na <sup>+</sup> /K <sup>+</sup> -ion pump. Transport and storage of dioxygen: active site structures and bio functions of O <sub>2</sub> -uptake proteins: hemoglobin, myoglobin, hemocyanin and hemerythrin, model synthetic dioxygen complexes. Electron transfer in biology (respiratory electron transport chain): active site structures and functions of cytochromes, cytochrome c. Iron-sulfur proteins: ferredoxines, rubredoxin, cytochrome c oxidase and model systems. | 12      |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>II</b>    | <b>Metalloenzymes</b><br>Copper enzymes, superoxide dismutase, cytochrome oxidase and ceruloplasmin; Coenzymes; molybdenum enzyme: xanthine oxidase; Nitrogenase and nitrogen fixation, zinc enzymes: carbonic anhydrase, carboxypeptidase and interchange ability of zinc and cobalt in enzymes; Vitamin B12 and B12 coenzymes.                                                                                                                              | <b>12</b> |
|              | <b>Chemistry of carbohydrates, proteins and nucleic acids</b><br>Chemistry of carbohydrates: Types of naturally occurring sugars: Deoxy sugar, amino sugar, branched chain sugar. Killiani-Fischer synthesis, Ruff's degradation, osazone formation, mutarotation.                                                                                                                                                                                            |           |
| <b>III</b>   | Metabolism of glucose. Bioenergetics: The ATP cycle. Proteins: Classification, Amino acid, property, peptide, general method of peptide synthesis, primary, secondary, tertiary and quaternary structure of protein. Determination of primary structure.<br>Nucleic Acids: DNA and RNA. Type of RNA and their function, Property of DNA in solution. Watson-Crick Model of DNA structure, replication, transcription and translation.                         | <b>12</b> |
| <b>IV</b>    | <b>Biochemistry of Iron</b><br>Structure and optical spectra; haeme proteins: magnetic susceptibility, epr and electronic spectra; haemoglobin and myoglobin: molecular structures, thermodynamics and kinetics of oxygenation, electronic and spatial structures, synthetic oxygen carriers, model systems; iron enzymes, peroxidase, catalase and cytochrome P-450; iron storage, transport, biomineralization and siderophores, ferritin and transferrins. | <b>12</b> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>48</b> |

#### Textbooks:

1. *Principles of biochemistry*; Lehninger A. L.; 6<sup>th</sup> edition; 2012; W. H. Freeman and company.
2. *Outlines of biochemistry*; Conn and Stumph; 5<sup>th</sup> edition; 1987; Wiley and sons, New York.
3. *Organic chemistry*; Solomon T. W. Graham; 12<sup>th</sup> edition; 2015; Wiley.
4. *Principles of Bioorganic Chemistry*, Lippard S.J.; Berg J.M.; 2nd edition, 2005; Panima Publ. Corp.

#### Reference books:

1. Palmer Trevor; *Understanding Enzymes*; 4<sup>th</sup> edition; 1995; Prentice Hall; UK.
2. Williams R. J. P. and Salvia F. R. De; *Biological chemistry of elements*; 2<sup>nd</sup> edition; 2001; Oxford University Press.

3. Kraatz H. & Metzler-Nolte N; *Concepts and Models in Bioinorganic Chemistry*, 3<sup>rd</sup> edition; 2006; Wiley.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

| SYLLABUS (3 <sup>rd</sup> SEMESTER)           |                 |                           |
|-----------------------------------------------|-----------------|---------------------------|
| Subject Name: Environmental & Green Chemistry | Level:500       | Subject Code: CHY014C303  |
| L-T-P-C-4-0-0-4                               | Credit Units: 4 | Scheme of Evaluation: (T) |

**Objective:** The objectives of **Environmental & Green Chemistry** are to provide the knowledge of major pollutants and different ways of controlling air, water, and soil pollution. The students will also get to know about the principles and methodologies of green chemistry

**Course Outcomes:**

| After successful completion of the course, students will be able to |                                                                                                            |                        |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                               | Course Outcome                                                                                             | Bloom's Taxonomy Level |
| CO1                                                                 | <b>Define</b> the major air pollutants and their mitigation methods                                        | BT1                    |
| CO2                                                                 | <b>Demonstrate</b> different water pollution and purification methods                                      | BT2                    |
| CO3                                                                 | <b>Apply</b> and <b>analyse</b> the knowledge of soil chemistry and solid waste management                 | BT3 & BT4              |
| CO4                                                                 | <b>Evaluate</b> the principles of green chemistry and <b>construct</b> new methodology for green synthesis | BT5 & BT6              |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Periods   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| I       | <b>Atmospheric Chemistry</b><br><br>Composition of atmosphere – major regions of atmosphere – Particles Ions and radicals in the atmosphere and their formation (formation of particulate matter, Ions and radicals), Air pollution – Major air pollutants – (Oxides of Carbon – Oxides of Nitrogen – Oxides of sulphur- Particulars – Smog and photochemical smog- Metallic pollutants –Radiation – Chemicals – Petroleum – Chlorofluorocarbons) – Effects of Air pollution (Acid rain, Green house effect, Global warming, Depletion of Ozone) –Control of air pollution. | 12        |
| II      | <b>Hydrosphere Chemistry</b><br><br>Distribution of chemical species in water, Gases in water, alkalinity, organic matter in water, criteria and standards of water quality- safe drinking water, Types of water pollutants – (Biological agents, Chemical agents, Physical agents), Toxic metals in water, Waste water treatment processes, Water purification for drinking and industrial purposes, disinfection techniques, demineralization and reverse osmosis.                                                                                                        |           |
| III     | <b>Soil Chemistry</b><br><br>Composition of soil, types of soil, Chemical properties – cation exchange capacity, $p^H$ , macro and micronutrients, Wastes from mining and metal production, Hazardous wastes and their disposal, Biodegradation of waste-anaerobic and aerobic treatment, Incineration, Pesticides and their role in the environment.                                                                                                                                                                                                                       | 12        |
| IV      | <b>Green Chemistry</b><br><br>Principles of green chemistry, principles of green organic synthesis, green alternatives of organic synthesis-coenzyme catalysed reactions, green alternatives of molecular rearrangements, electrophilic aromatic substitution reactions, oxidation-reduction reactions, clay catalysed synthesis, condensation reactions, Green photochemical reactions, Green Solvents, Introduction to microwave assisted reactions.                                                                                                                      | 12        |
|         | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>48</b> |

**Text books:**

1. *Air Pollution: its Origin and Control*; K. Wark, C. F. Warner & W. T. Davis, 3<sup>rd</sup> edition, 1997, Pearson
2. *Environmental Pollution*, A.K. De, 6<sup>th</sup> edition, 2006, New Age International, New Delhi.
3. *Environmental Chemistry*, B.K. Sharma & H. Kaur, 2<sup>nd</sup> edition, 2003, Goel Publishing house, Meerut

**Reference books**

1. Rao C.S., *Environmental Pollution Control Engineering*, 2<sup>nd</sup> edition, 2006, New Age International
2. Sanghi R. and Srivastava M. M., *Green Chemistry: Environment Friendly alternatives*, 2<sup>nd</sup> edition, 2008, Narosa Publishing House, New Delhi, India.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

**SYLLABUS (3<sup>rd</sup> SEMESTER)**

**Subject Name: Physical Chemistry Lab      Level: 400      Subject Code: CHY014C314**

**L-T-P-C – 0-0-8-4**

**Credit Units: 4**

**Scheme of Evaluation: (P)**

**Objectives:** The objective of Physical Chemistry Laboratory is to provide the knowledge of designing particular experiment in both theoretical and experimental aspects, to develop curiosity and interest as well as to improve the understanding of the concepts in analyzing various physical phenomenon in chemical processes.

**Course Outcomes**

| After successful completion of the course, student will be able to |                                                                                                                                                                                                                                                     |                        |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl. No.                                                            | Course Outcome                                                                                                                                                                                                                                      | Bloom's Taxonomy Level |
| 1                                                                  | Be able to handle different instruments important in physical and material sciences.                                                                                                                                                                | BT 1                   |
| 2                                                                  | Be able to get detailed concepts of kinetics of different reactions, autocatalytic reaction and volumetric chemical analysis by doing iodometric titration and able to understand the applications of conductivity, pH-metry and spectrophotometry. | BT 2                   |
| 3                                                                  | Apply previous knowledge in analyzing the experimental data to get conclusions.                                                                                                                                                                     | BT3 & BT4              |
| 4                                                                  | Design the experiment, collect the data and solve the problem with independent thinking.                                                                                                                                                            | BT5 & BT6              |

## **Detailed Syllabus:**

*Note:* Students shall complete 14 experiments of 7 instrumental and 7 non-instrumental. In the semester end examination, students shall perform 2 experiments.

### **Unit 1: Chemical kinetics**

- 1) Determine the temperature coefficient and energy of activation of acid hydrolysis of methyl acetate, using least-square calculation.
- 2) Study the kinetics of the reaction between iodine and acetone in acidic medium by half-life period method and determine the order with respect to iodine and acetone.
- 3) Study the saponification of ethyl acetate by sodium hydroxide and determine the order of the reaction and energy of activation.
- 4) Study the autocatalytic reaction between oxalic acid and  $\text{KMnO}_4$  and determine the order of the reaction.
- 5) Determine the inversion of sucrose in presence of two acids polarimetrically using Guggenheim plots and hence determine the relative strengths of the acids.

### **Unit 2: Conductometry**

- 1) Determine the equivalent conductivity of acetic acid at infinite dilution by Kohlrausch's method and hence find out the degree of dissociation of the acid.
- 2) Find out the relative strength of acetic acid and monochloroacetic acid by conductance measurement.
- 2) Determine the strength of the components of the following mixtures by conductometric titration.
  - (a) Hydrochloric acid and acetic acid.
  - (b) Sulphuric acid and copper sulfate.

### **Unit 3: pH-metry and potentiometry**

- 1) Determine the dissociation constant of acetic acid/ oxalic acid using Hendersen's equation.
- 2) Find the amount of the components of the following mixtures using pH – metric titration.
  - a) Hydrochloric acid + acetic acid
  - b) Hydrochloric acid + oxalic acid

### **Unit 4: Spectrophotometry**

- 1) Verify Beer's law and determine the concentration of solutions like  $\text{KMnO}_4$  /  $\text{K}_2\text{Cr}_2\text{O}_7$  /  $\text{CuSO}_4$
- 2) Determine the composition of iron-thiocyanate complex spectrophotometrically by Job's method of continuous variation.

### **Unit 5: Miscellaneous experiments**

- 1) Determine the molar mass of a polymer by viscometry method.
- 2) Perform theoretical calculations using a computer on potential energy diagram of hydrogen molecule ion.
- 3) Determine the coefficient of viscosity of a liquid by Ostwald's viscometer.
- 4) Determine the surface tension of a liquid by Stalagmometer.

**Text Book:**

1. *Advanced Practical Physical Chemistry*; Yadav, J.B.; 28<sup>th</sup> edition; 2009; Goel Publishing House

**Reference Books:**

1. Gurtu, J.N., Gurtu, A.; *Advanced Physical Chemistry Experiments*, 6<sup>th</sup> edition, 2014, Pragati Prakashan
2. Halpern, M.; *Experimental Physical Chemistry*, 2<sup>nd</sup> edition, 1988, Prentice Hall, Upper Saddle River, NJ 07458

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

|                                           |
|-------------------------------------------|
| <b>SYLLABUS (4<sup>th</sup> SEMESTER)</b> |
|-------------------------------------------|

|                                      |                        |                                 |
|--------------------------------------|------------------------|---------------------------------|
| <b>Subject Name: Spectroscopy-II</b> | <b>Level: 500</b>      | <b>Subject Code: CHY014C401</b> |
| <b>L-T-P-C – 4-0-0-4</b>             | <b>Credit Units: 4</b> | <b>Scheme of Evaluation: T</b>  |

**Objectives:** The objectives of **Spectroscopy-II** are

- To provide a thorough background on those that are particularly important to analytical chemistry
- To provide knowledge, techniques, and principles of spectroscopy and to develop the ability to apply skills in the proper handling of apparatus
- To develop problem-solving skills

**Course Outcomes:**

| After successful completion of the course, students will be able to |                                                                                                                                                    |                        |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| SI No                                                               | Course Outcome                                                                                                                                     | Bloom's Taxonomy Level |
| CO1                                                                 | <b>Define</b> and get a detailed idea of rotational, vibrational, and Raman spectroscopy, as well as their molecular geometry and selection rules. | BT1                    |
| CO2                                                                 | <b>Explain</b> the origin of chirality, spectroscopic techniques for macromolecules, and their applications                                        | BT2                    |
| CO3                                                                 | <b>Apply</b> the concept of Mössbauer spectroscopy and ESR spectroscopy, and their applications                                                    | BT3                    |
| CO4                                                                 | <b>To get information</b> about NMR and Electronic spectroscopy in inorganic chemistry                                                             | BT4                    |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Periods |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <p><b>Rotational and Vibrational spectroscopy</b></p> <p>Rotational spectroscopy: Classification of molecules based on their moment of inertia, rotational energy levels, molecular geometry determination, Stark effect, molecular dipole moment. Rotational spectroscopy of symmetric and asymmetric top molecules.</p> <p>Vibrational spectroscopy: Harmonic and anharmonic oscillators. Morse potential, mechanical and electrical anharmonicity, selection rules. The determination of anharmonicity constant and equilibrium vibrational frequency from fundamental and overtones. Vibrational selection rules using symmetry, polarization of transitions. Normal modes analysis using group theory.</p> <p>Raman spectroscopy- polarizability tensor, Stokes and anti-Stokes lines, instrumentation and applications in chemical and biological systems.</p>                                                                            | 12      |
| II      | <p><b>Principles behind CD/ORD spectroscopy and application</b></p> <p>CD/ORD: symmetry origin of optical activity of molecules. Phenomenon of Optical Rotatory Dispersion (ORD) and Circular Dichroism (CD): principle, methodology and applications, molecular dissymmetry and chiroptical properties, Cotton effect, Faraday effect in magnetic circular dichroism. Application of CD/ORD spectroscopy for the study of metal-ligand equilibria</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 12      |
| III     | <p><b>Theory of Mössbauer and Electron Spin Resonance (ESR) spectroscopy with its application</b></p> <p>Mössbauer spectroscopy: Gamma ray emission and absorption by nuclei, Mössbauer effect, conditions, nuclear recoil, Doppler effect, instrumentation, chemical shift, quadrupole effect, effect of magnetic field, effect of simultaneous magnetic and electric fields.</p> <p>Electron Spin Resonance (ESR) spectroscopy: Introduction, behavior of a free electron in an external magnetic field, basic principle, hyperfine coupling in isotropic system. Factors affecting magnitude of g-values. Line width, double resonance.</p> <p>Application of Mössbauer spectroscopy to the study of high-spin and low-spin iron compounds and in coordination complexes.</p> <p>Application of ESR spectroscopy in transition metal complexes having one unpaired electron including biological systems and to inorganic free radicals.</p> | 12      |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>IV</b>    | <b>NMR and Electronic spectroscopy in inorganic chemistry</b><br>Fluorescence and phosphorescence spectroscopy: Jablonski Diagram, origin of fluorescence and phosphorescence processes, quantum yield, fluorescence quenching-static and dynamic. Instrumentation and applications.<br>NMR spectroscopy: Simple application to diamagnetic inorganic compounds; NMR paramagnetic shifts, simple application to paramagnetic compounds; NMR of $^{31}\text{P}$ and $^{19}\text{F}$ in inorganic compounds.<br>Photoelectron spectroscopy: Basic principles and applications of PES ( $\text{O}_2$ , $\text{N}_2$ and $\text{N}_3^-$ only); chemical information from ESCA. | <b>12</b> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>48</b> |

**Text books:**

1. *Fundamentals of molecular spectroscopy*; Banwell Colin N., McCashEllain M; 4<sup>th</sup> edition; 2001; Tata Macgraw-Hill.

**Reference books:**

1. Rao C. N. R. and Ferraro J. R.; *Spectroscopy in Inorganic Chemistry*; Vol. I & II; 1970 and 1971; Wiley; New York.
2. Greenwood N. N. and Gibb T. C.; *Mossbauer spectroscopy*; 1977; Chappman and Hall ltd; London.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

| SYLLABUS (4 <sup>th</sup> SEMESTER)           |                 |                          |
|-----------------------------------------------|-----------------|--------------------------|
| Subject Name: Chemical Kinetics and Catalysis | Level: 500      | Subject Code: CHY014C402 |
| L-T-P-C – 4-0-0-4                             | Credit Units: 4 | Scheme of Evaluation: T  |

**Objective:** The objectives of **Chemical Kinetics and Catalysis** are

- To study the details of the kinetics of fast reactions, unimolecular reactions, and reactions in solutions
- To acquire knowledge of photochemical reactions and kinetics of different photochemical and electrochemical reactions.
- To understand the theories of heterogeneous catalysis and the properties of zeolites and clays.

**Course Outcomes:**

| After successful completion of the course, students will be able to |                                                                                    |                        |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------|
| Sl No                                                               | Course Outcome                                                                     | Bloom's Taxonomy Level |
| CO1                                                                 | Be able to understand the kinetics of unimolecular reactions.                      | BT 1                   |
| CO2                                                                 | Be able to understand different techniques used to study fast reactions.           | BT 2                   |
| CO3                                                                 | Be able to understand the properties of zeolites and clays                         | BT 3                   |
| CO4                                                                 | Be able to acquire knowledge of different photochemical reactions.                 | BT 4                   |
| CO5                                                                 | Know the theories of heterogenous catalysis and kinetics of reactions in solutions | BT 5                   |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Periods |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <b>Chemical Kinetics</b><br><i>Study of fast reactions:</i> Stopped flow technique, temperature and pressure jump methods. NMR studies in fast reactions, shock tube kinetics, relaxation kinetics. relaxation time in single step fast reactions, determination of relaxation time.<br><i>Theories of unimolecular reactions:</i> Limitations of Hinshelwood's treatment, RRK theory, Slater's treatment, RRKM theory.<br><i>Kinetics of reactions in solution:</i> Diffusion controlled, TST of reactions in solution, Bronsted and Bjerrum equation, effect of ionic strength, kinetic salt effect. Kinetics of electrode reactions: Butler-Volmer equation, Tafel Plots. | 10      |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| II           | <b>Photochemical reactions</b><br><i>Photochemical reactions:</i> Photophysical kinetics- state energy diagrams. Delayed fluorescence- the mechanism and kinetics of fluorescence quenching – Stern-Volmer equation.<br><i>Chemical kinetics in the elucidation of reaction mechanism:</i> Reaction in compounds containing carbonyl groups: photo reduction and related reaction, photocycloaddition reactions, transition metal complexes.                                                                                                                  | 10        |
| III          | <b>Zeolites and Clays</b><br>Zeolites (natural and synthetic)- shape selectivity properties- solid acids, acidity of zeolites and clays. Mesoporous materials, poorly crystalline silicates and aluminosilicates-MCM-41 type materials. Applications of zeolites and clays as heterogeneous catalysts in cracking, reforming, and olefin reactions. Zeolites as catalyst supports.<br><i>Surface Characterization Techniques:</i> Ultra-high vacuum for surface studies, Auger electron spectroscopy, Photoelectron spectroscopy, Scanning probe microscopy,. | 10        |
| IV           | <b>Kinetics of Heterogeneous Catalysis</b><br>Surface area determination from adsorption isotherms and point-B methods, porosity determination by volumetric and gravimetric methods. Chemisorption on metals, semi-conducting oxides, and insulator oxides.<br>Kinetics of heterogeneous catalysis, effect of temperature on rates of catalyzed reactions, Langmuir–Hinshelwood and Eley–Rideal mechanisms, mass transport limitation of catalyzed reactions.<br>Surface dependence of reaction rates, volcano principles                                    | 10        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>40</b> |

### **Text Books:**

1. *Chemical Kinetics*; Laidler, K.J.; 3<sup>rd</sup> Edition; 2012; Pearson
2. *Fundamental of Photochemistry*; Mukherjee-Rohatgi, K.K.; 3<sup>rd</sup> edition, 2014, New age international (P)Ltd
3. *Electrochemical Methods: Fundamental and Applications*; J.B. Allen and Faulkner, L.R.; 2<sup>nd</sup> edition, 2000, Wiley
4. *Heterogeneous Catalysis: Principles & Applications*, Bond, G.C.; 2<sup>nd</sup> edition, 1987, Oxford University Press
5. *Physical Chemistry of Surfaces*; Adamson, A.W. and Gast, A.P.; 6<sup>th</sup> edition; 1997; John Wiley and Sons, Canada

### **Reference Books:**

1. Atkins P. W. and Paula J. de; *Physical Chemistry*; 10<sup>th</sup> edition; 2014; Oxford University Press
2. Levine, I.; *Physical Chemistry*; 6<sup>th</sup> edition; 2011; Tata McGraw Hill.
3. Puri, B.R.; Sharma, L.R.; Pathania, M.S.; *Principles of Physical Chemistry*; 47<sup>th</sup> edition; 2016; Vishal Publishing Company
4. J.O. Bockris, A. K. N. Reddy; *Modern Electrochemistry Part 1, 2A and 2B*; 2<sup>nd</sup> Edition, Springer

- Chakrabarty, D.K. and Viswanathan, B.; *Heterogeneous Catalysis*; 1<sup>st</sup> edition; 2011; New Age International (P)Limited
- Thomas, M.; Thomas, W.J.; *Introduction to principles of heterogeneous catalysis*; 1<sup>st</sup> edition, 1967; Academic Press, New York
- Somorjai, G.A.; *Introduction to surface chemistry and catalysis*, 2<sup>nd</sup> edition; 2010; Wiley-Blackwell,

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

### SYLLABUS (4<sup>th</sup> SEMESTER)

|                                                       |                 |                          |
|-------------------------------------------------------|-----------------|--------------------------|
| Subject: Heterocyclic Compounds & Medicinal Chemistry | Level:500       | Subject Code: CHY014C403 |
| L-T-P-C – 4-0-0-4                                     | Credit Units: 4 | Scheme of Evaluation: T  |

**Objective:** The objectives of **Heterocyclic compounds & Medicinal Chemistry** are

- To help students in the development of curiosity and interest in Medicinal Chemistry
- To help students to understand the structure and reactions of biological molecules
- To provide the students the fundamental concepts required to rationalise and predict the structure of an unknown drug

#### **Course Outcomes:**

| After successful completion of the course, student will be able to |                                                                                                          |                        |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                              | Course Outcome                                                                                           | Bloom's Taxonomy Level |
| CO1                                                                | <b>Understand</b> the classification, nomenclature, and chemical properties of heterocyclic compounds.   | BT1                    |
| CO2                                                                | <b>Explain</b> the methods of synthesis and chemical reactivity of heterocycles.                         | BT2                    |
| CO3                                                                | <b>Apply</b> the knowledge of medicinal chemistry develop curiosity and interest in Medicinal Chemistry. | BT3                    |
| CO4                                                                | <b>Analyze</b> the synthetic pathways for important therapeutic agents.                                  | BT4                    |

**Detailed Syllabus:**

| Modules | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Periods |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| I       | <p><b>Quinoline and Isoquinoline:</b></p> <p>Synthesis of quinolines from anilines, from <i>ortho</i>-aminoaryl ketones or aldehydes, synthesis of isoquinolines from 2-arylethamines, from aryl-aldehydes and an aminoacetaldehyde acetal, from <i>ortho</i>-alkynyl aryl-aldehydes or corresponding imines, electrophilic substitution reaction of Quinoline and Isoquinoline.</p> <p><b>Indole:</b></p> <p>Synthesis of indoles from arylhydrazones, from <i>ortho</i>-nitrotoluenes, from <i>ortho</i>-aminoaryl alkynes, from <i>ortho</i>-alkylaryl isocyanides, from <i>ortho</i>-acyl anilides, electrophilic substitution reaction of Indole.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 | 12      |
| II      | <p><b>Basic Chemistry of Alkaloids, Terpenoids and Steroids</b></p> <p>Source, structural types of alkaloids, classification, structure elucidation, reactions and synthesis of Nicotine and Papaverine. Reactions and synthesis of Quinine and Morphine.</p> <p>Isoprene rule, general introduction to sesqui-, di- and tri- terpenoids, structure elucidation and synthesis of representative examples of acyclic, monocyclic and bicyclic monoterpenes.</p> <p>Reaction and synthesis of Steroids: Cholesterol, Bile acid, Testosterone, Estrone, Progesterone.</p> <p>Structure and synthesis of Prostaglandins: PGE<sub>2</sub>, PGF<sub>2α</sub>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                | 12      |
| III     | <p><b>Drug Discovery and Design</b></p> <p>Design and development of a drug: Choosing a disease, choosing a drug target, target specificity and selectivity, multi-target drugs.</p> <p>Identifying a bio-assay: Choice of bioassay, <i>in vitro</i> and <i>in vivo</i> tests, high through put screening.</p> <p>Finding a lead compound: Screening of natural products, synthetic compound library and existing drugs, combinatorial and parallel synthesis, computer-aided design of lead compounds, serendipity and the prepared mind; isolation and purification.</p> <p>Optimizing target interactions: Structure-activity relationship-binding role of different organic functional groups, identification of a pharmacophore.</p> <p>Different strategies in drug design: Variation of substituents, extension of the structure, chain extension/contraction, ring expansion/contraction, isosteres and bioisosteres, simplification of the structure, rigidification of the structure, conformational blockers.</p> <p>Prodrugs: Different roles of prodrugs.</p> | 12      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| IV | <b>Study of Antibacterial Agents and Anti-cancer Agents</b><br>History of Antibacterial agents, mechanism of action. Sulphonamides (Antimetabolite): History, mechanism of action, applications.<br>Penicillin: History, structure and properties of benzylpenicillin, mechanism of action of penicillin, resistance to penicillin, methods of synthesizing penicillin analogues.<br>Anticancer agents: An introduction, causes of cancer, genetic faults leading to cancer-protooncogenes and oncogenes, abnormal signaling pathways, insensitivity to growth-inhibitory signals, abnormalities in cell cycle regulation, telomeres, angiogenesis, tissue invasion, and metastasis.<br>Treatment of cancer: Drugs acting directly on nucleic acids- intercalating agents, non-intercalating agents which inhibit the action of topoisomerase enzymes on DNA, alkylating and metalating agents (Nitrogen mustards, Nitrosoureas, Busulfan, Cisplatin, Dacarbazine and Procarbazine, Mitomycin). | 12        |
|    | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>48</b> |

#### **Text Books:**

1. *An Introduction to Medicinal Chemistry*; Patric G. L.; 6<sup>th</sup> edition; 2017; Oxford University Press.
2. *Fundamental of Medicinal Chemistry*; Thomas G.; 2<sup>nd</sup> edition, 2007; Wiley.
3. *Heterocyclic Chemistry*; Joule J. A., Mills K.; 5th Edition; Jun 2010; Wiley-Blackwell.
4. *Principles of Modern Heterocyclic Chemistry*; Paquette L. A.; Jan 2010; University of Minnesota.

#### **Reference Books:**

1. King F. D.; *Medicinal Chemistry: Principles and Practice*; 2<sup>nd</sup> edition; Royal Society of Chemistry
2. Clayden J., Greeves N. and Warren S., *Organic chemistry*; 2<sup>nd</sup> edition; Oxford University press.

| Credit Distribution |           |                       |
|---------------------|-----------|-----------------------|
| Theory/ Tutorial    | Practicum | Experimental Learning |
| 72                  | ---       | 48                    |

### SYLLABUS (4<sup>th</sup> SEMESTER)

**Subject: Organometallic Chemistry and Catalysis    Level: 500    Subject Code: CHY014C404**  
**L-T-P-C – 4-0-0-4                      Credit Units: 4                      Scheme of Evaluation: (T)**

The objective of **Organometallic Chemistry** is to provide detailed understanding of the main group organometallic compounds. To cover the idea of application of organometallic compounds to organic synthesis and catalysis.

**Course Outcomes:**

| After successful completion of the course, the students will be able to |                                                                                                                                         |                        |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sl No                                                                   | Course Outcome                                                                                                                          | Bloom's Taxonomy Level |
| CO1                                                                     | <b>Define</b> and get the detailed idea synthesis, structure and reactivity of main group organometallic compounds                      | BT1                    |
| CO2                                                                     | <b>Explain</b> the problems related to their structure and bonding of the various organometallic compounds and predict its stability.   | BT2                    |
| CO3                                                                     | <b>Apply</b> the idea of organometallic chemistry to understand th syntheses and reactions of cyclopentadienyl and arene meta analogues | BT3                    |
| CO4                                                                     | <b>Analyse</b> the catalytic mechanism of different industrially important organic synthesis.                                           | BT4                    |

**Detailed Syllabus:**

| Modules    | Topics & Course content                                                                                                                                                                                                                                                                                                                                                                           | Periods   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>I</b>   | <b>Main Group Organometallics</b><br>Synthesis and reactions of organolithium compounds; Synthesis and reactions of organomagnesium compounds; Organometallics of zinc and mercury: preparation, structure, bonding and reactions of aluminum organyls; Thallium(I) organyls (synthesis of TlCp); Organyls of sodium, synthesis of NaCp; Silicon and tin organyls of coordination number 4.       | <b>12</b> |
| <b>II</b>  | <b>Transition Metal–Carbon Bond</b><br>Transition Metal–Carbon $\sigma$ -Bond: Brief review of metal alkyl compounds; transition metal carbene and transition metal-carbyne compounds; transition metal vinylidene and transition metal allenylidene compounds.<br>Transition Metal-Carbon $\pi$ -Bond: Cyclopropenyl cation ( $C_3R_3^+$ ) as a ligand; $C_4R_4$ as a ligand ( $R = H, Me, Ph$ ) | <b>12</b> |
| <b>III</b> | <b>Syntheses and reactions of Cyclopentadienyl and Arene Metal Analogues</b><br>Synthesis and reactions of cyclopentadienyl metal carbonyls, cyclopentadienyl metal hydrides, cyclopentadienyl metal halides, arene metal carbonyls, $\eta^6$ -arene-chromium tricarbonyl in organic synthesis.                                                                                                   | <b>12</b> |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>IV</b>    | <b>Applications to Organic Synthesis and Homogeneous Catalysis</b><br><b>In Organic Synthesis:</b> Hydrozirconation of alkenes and alkynes; Carbonylation of Colman's reagent; $\eta^4$ -diene iron-tricarbonyls in organic synthesis<br><b>In Catalysis:</b> Asymmetric hydrogenation; synthesis of acetic acid and glycol (Monsanto acetic acid process); arylation/vinylation of olefins (Heck reaction); Wacker process (olefin oxidation); Asymmetric epoxidation | <b>12</b> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>48</b> |

**Text Books:**

1. *Organometallics*; Elschenbroich.C; 3<sup>rd</sup> edn.,2006; Wiley-VCH Publication.
2. *Advanced Inorganic Chemistry*; Cotton F. A. & Wilkinson. G.; 5<sup>th</sup>edn.; 1988, John Wiley

**Reference Books:**

1. Crabtree, R. H.; *The Organometallic Chemistry of the Transition Metals*; 4<sup>th</sup> edn.; 2005; John Wiley.
2. Bochmann.M; *Organometallics-I Complexes with Transition Metal- Carbon  $\sigma$ -Bonds*, 3<sup>rd</sup> edition, 1994; Oxford Chemistry Primers.
3. Bochmann.M; *Organometallics-2 Complexes with Transition Metal– Carbon  $\pi$ -bonds*; 3<sup>rd</sup> edition, 1994; Oxford Chemistry Primers.

| <b>Credit Distribution</b> |                  |                              |
|----------------------------|------------------|------------------------------|
| <b>Theory/ Tutorial</b>    | <b>Practicum</b> | <b>Experimental Learning</b> |
| <b>72</b>                  | <b>---</b>       | <b>48</b>                    |